Abdullah Khaled

Undergraduate applicant — electrical engineering & Islamic studies; embedded systems, vision, robotics

Frisco, TX   me@aakhaled.com   www.aakhaled.com   akhaled247   akhaled247

Education

Frisco ISD Career and Technical Education Center, Project Lead The Way – Frisco, TX — Aug 2023 – May 2027

- Frisco ISD Career and Technical Education Center Project Lead the Way (PLTW) Nationally Distinguished Engineering Program
- Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
- Engineering Portfolio: aakhaled.com/pltw-engineering

Wakeland High School, High School Diploma – Frisco, TX — Aug 2023 – May 2027

- Unweighted GPA: 4.0
- 18 AP courses by graduation (completed and planned)

Engineering Projects

OralVision (Oral Disease Detector) – Conrad Challenge; Diamond Challenge — Oct 2025 – present

- Developed privacy-preserving oral disease screening pipeline (centrally trained model + on-device inference, without transmitting raw patient images), with my engineering focus on the imaging device end-to-end while my partner focused on the inference/training layer
- Designed, wired, and programmed the imaging device (XIAO-ESP32-C6 MCU + OV5642 5MP camera), including power-aware capture (camera current draw can trip the 5V voltage regulator)
- Bench-tested image capture quality and robustness across lighting/viewing angles to maximize reliable low-cost imaging
- Verified repeatability by running multiple capture sessions under comparable conditions and confirming output consistency
- Iterated through 3 PCB revisions and 5 3D-printed snap-fit case revisions to improve reliability and manufacturability
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Conrad Challenge Innovator Award
- Diamond Challenge Semifinalist
- Patent application in progress with United States Patent and Trademark Office (USPTO)

WatchFall (Senior Fall Monitor Device) – Samsung Solve for Tomorrow — Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate
- Integrated Arducam HM01B0 Monochrome QVGA SPI Camera Module with Raspberry Pi Pico for resource-aware frame capture
- Planned next validation: evaluate false positives and alert responsiveness on recorded motion/fall scenarios, then tune fall rules and thresholds before expanded on-device testing

Work Experience

Math and Reading Instructor, Gideon Math and Reading Center – Frisco, TX — June 2024 – present

- Taught groups of ~4–12 students/day from kindergarten to 11th grade in math, reading, and grammar (including Algebra and Geometry), using guided learning sessions
- Analyze student performance data with guided checkpoints to identify learning gaps and develop personalized plans
- Evaluate reading material difficulty and adapt instruction to match target comprehension levels using targeted guided practice

Math and Reading Teaching Assistant (TA), Gideon Math and Reading Center – Frisco, TX Jan 2022 – May 2024

- Graded students' work from kindergarten to 11th-grade, including Algebra and Geometry
- Aided students in completing their assignments appropriate to their age and skill level
- Increased grading efficiency by 50%

Technical and Computer Skills

Engineering: CAD/CAM: Fusion 360 (Autodesk), Onshape; NI Multisim, KiCAD, EasyEDA, Soldering

Technologies: WPILib, FTC SDK, OpenCV, SolvePnP, TensorFlow, Federated Learning, Kalman Filtering, Finite State Machines, Mechanism Simulation, Control Theory

Software: Python, Java (robotics), JavaScript, HTML, Jekyll, CSS, Google Workspace, R, Google Sheets, MS Office, Git, GitHub

Extracurricular Activities

Programming Lead, ITKAN Robotics - FIRST Robotics Competition (FRC 9128) – Plano, TX Sept 2024 – Feb 2026

- Lead FRC 9128 robot software for a ~50-member team; as Programming Lead (2024-12–2025-06) owned ~80% of the codebase as primary Java/WPILib developer, maintaining GitHub workflows and command-based autonomous and teleop systems
- Integrated OpenCV + SolvePnP and tuned Kalman fusion between vision and drivetrain state; shipped autonomous pathing and targeting (state machines, Bezier-based planning, PID), improving average autonomous points per match by ~50% across 12 matches after vision integration
- Implemented driver-input buffering for teleop handoff to automated placement; built match logging/debug tooling and simulation workflows to validate behavior during an 8-week build season with limited robot access; taught mechanism simulation to supporting students
- Mentored rookie FTC teams 26357 and 28931 (2024–2025); supported ~30 students (6 programmers) from prototyping through competition, including ~25 hrs/week at peak ITKAN FTC build season
- Applied FRC sensor-fusion experience to a custom FTC pose estimator (wheel odometry, IMU, vision) with Kalman filtering for localization and automated turret targeting; improved targeting/scoring accuracy by ~35%; FTC codebase
- Delivered three 1-hour ICF FLL workshops (7 students/session) for refugee students using the self-developed OARobotics curriculum (OARobotics); students built FLL-class robots and completed block-code autonomous maze navigation by session end

Chief Technical Advisor, ITKAN Robotics - FIRST Robotics Competition (FRC 9128) – Plano, TX July 2025 – present

- Lead software strategy for ITKAN's multi-team FRC program (~20 programmers, ~100-member organization), delegating work, standardizing codebase layout, and aligning autonomy and teleop with mechanical/electrical integration under build-season deadlines
- Managed autonomy and teleop reliability engineering, implementing layered failsafes to meet >90% reliability expectations despite limited build-season time and staffing; coordinated cross-functional integration so contingency planning preserved match performance after a near-failure camera disconnect
- Improved mechanism cycle time from ~2 s to <1 s (intake-to-top) and increased shooter accuracy from ~70% to ~95%, doubling throughput from ~10 to ~20 balls/s (lab-measured before/after under comparable conditions)
- Completed autonomous routines without error in ~8–10 unofficial practice runs (pre-competition testing)
- Mentored programming students and led sponsor and university/community communication, including Toyota and U.S. Senator Chris Van Hollen at an in-person event
- Built Mantik (Java + WPILib training website) and produced an FRC programming tutorial playlist (YouTube series); Google Search Console reach over a 2-month period reached 50+ students in the United States
- Supported multi-team FRC outcomes including FRC 9128 Texas State Championship playoff qualification; program results on The Blue Alliance (9128) and 10340
- Ranked highly on Statbotics for Highest Combined Clean Scores among global FRC teams (snapshot Mar 2026); https://www.statbotics.io/matches

Founder and President, Wolverine Robotics — Wakeland High School (FTC 33791) – Frisco, TX — Aug 2024 – present

- Founded school FTC program (team 33791) and led Wolverine Robotics to an undefeated run and 3rd place at the UIL State Championship; grew membership from 0 to 50+ in year one (~75 roster in year two; 25–50% active participants)
- Secured ~$6,000+ in resources and built a competition-ready FTC robot in under two weeks (approx. $2,000 school allocation + $2,000 NHS fundraising + $2,000 parent donations)
- Presented club vision and resourcing plan to school administration (principal, assistant principal/club manager, physics teacher sponsor) to obtain approval, a teacher sponsor, and permission to compete
- Delivered ~6 technical workshops (10–15 attendees) on assembly, Java, FTC simulations, and Onshape CAD; created onboarding presentations and an engineering notebook; led monthly meetings with robot demonstrations
- Implemented FTC autonomous/pathing (FTC Libraries, PedroPathing) with dead-reckoning wheel odometry and IMU fusion for real-time localization

Teaching Assistant (TA), Islamic Center of Frisco (ICF) – Frisco, TX — Sept 2023 – May 2024

- Aided elementary students in religious studies and helped keep them on task and engaged
- Assisted Primary Instructor with curriculum implementation

Co-Founder, Secretary, Muslim Student Association (MSA), Wakeland High School – Frisco, TX — Aug 2024 – present

- Developed mission statement, recruited teacher sponsor and members, and presented activities to school administration
- Coordinated with school staff to establish a prayer room for weekly prayers
- Created various media for promoting the club and prayer room, including social media posts

Tutoring Program Coordinator, Ma'ruf Dallas Refugee Organization (Project Taleem) – Dallas, TX — Sept 2023 – present

- Co-lead Project Taleem (~2 hr/week during the school year; pauses for summer and winter breaks), tutoring K–6 refugee students in small groups (~1:2–3), typically 3–10 learners per session alongside ~4–5 peer instructors
- Coordinate Ma'ruf and Gideon organizers to supply discounted Gideon learning materials; place students with IXL math and reading diagnostics spanning phonics through ~3rd-grade reading passages and addition through fractions
- Lead pre-session briefings for peer instructors; assign age-banded groups and smaller ratios for newer volunteers; supervise instructional delivery while maintaining a direct tutoring load; train peers on Gideon-based methods informed by parallel Gideon student, TA, and instructor roles
- Recruit high-school volunteers from five schools via robotics networks and onboard them into the program; provide periodic updates to the volunteer coordinator
- Run multilingual sessions with peer interpretation when available, mid-session breaks, and age-separated groupings to stabilize pacing and engagement
- Deliver monthly robotics enrichment for Project Taleem using the OARobotics curriculum (www.aakhaled.com/personal/oarobotics)
- Measured average ~15-point IXL SmartScore increase over four months (Nov–Feb; n=10 students)

Vice President, Mu Alpha Theta Math Honor Society, Wakeland High School – Frisco, TX — Aug 2024 – present

- Organized and designed presentations for meetings
- Expanded Math Lab Program and coordinated peer tutoring, attaining over 20% member participation
- Organized annual middle school math competition with 100+ participants, 3 participating schools, and 35+ volunteers to promote STEM to middle school students

Treasurer, Mu Alpha Theta Math Honor Society, Wakeland High School – Frisco, TX — Aug 2023 – July 2024

- Designed and implemented Math Lab Program, coordinating student-led tutoring in Algebra I courses with teachers
- Built website for Math Lab Program using Google Sites for volunteer sign-up
- Managed budget for 100+ members

Volunteer, Frisco Family Services – Frisco, TX — Sept 2023 – present

- Sorted and shelved donations for Frisco Family Market, which raises funds for underserved local families with financial, medical, and physical needs
- Sorted and stocked Food Pantry, which also serves underserved local families

Junior Varsity (JV), Wolverines Tennis Team, Wakeland High School – Frisco, TX — Aug 2023 – Aug 2024

- Participated in singles and doubles tournaments throughout academic school year
- Sportsmanship Award Winner 2024

Honors & Awards

- (2025) Collegeboard: AP Scholar with Distinction, National Recognition Program: School Recognition Award
- (2026) OralVision: Conrad Challenge Innovator Award; Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner, Control Award Winner (FiT-North Dallas Semi-Regional Championship, 25 teams), Control Award Winner (FiT-North U-League Tournament)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist, Event Winner (FIRST® in Texas Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)